

Finger Pose Estimation for Under-Screen Fingerprint Sensor

Xiongjun Guan^{ID}, Zhiyu Pan^{ID}, Jianjiang Feng^{ID}, *Member, IEEE*, and Jie Zhou^{ID}, *Fellow, IEEE*

Abstract—Two-dimensional pose estimation plays a crucial role in fingerprint recognition by facilitating global alignment and reduce pose-induced variations. However, existing methods are still unsatisfactory when handling with large angle or small area inputs. These limitations are particularly pronounced on fingerprints captured by under-screen fingerprint sensors in smartphones. In this paper, we present a novel dual-modal input based network for under-screen fingerprint pose estimation. Our approach effectively integrates two distinct yet complementary modalities: texture details extracted from ridge patches through the under-screen fingerprint sensor, and rough contours derived from capacitive images obtained via the touch screen. This collaborative integration endows our network with more comprehensive and discriminative information, substantially improving the accuracy and stability of pose estimation. A decoupled probability distribution prediction task is designed, instead of the traditional supervised forms of numerical regression or heatmap voting, to facilitate the training process. Additionally, we incorporate a Mixture of Experts (MoE) based feature fusion mechanism and a relationship driven cross-domain knowledge transfer strategy to further strengthen feature extraction and fusion capabilities. Extensive experiments are conducted on several public datasets and two private datasets. The results indicate that our method is significantly superior to previous state-of-the-art (SOTA) methods and remarkably boosts the recognition ability of fingerprint recognition algorithms. Our code is available at <https://github.com/XiongjunGuan/DRACO>

Index Terms—Fingerprint, pose estimation, fingerprint recognition, multimodal, decoupled probability distribution, feature fusion, knowledge distillation, knowledge transfer.

I. INTRODUCTION

TWO-DIMENSIONAL pose estimation has been extensively researched in the field of fingerprint [1], [2], [3], [4], [5], [6], [7], [8], [9], [10]. This task aims to determine the fingerprint's center position and rotation direction from an input image, enabling the effective alignment of heterogeneous data within a unified coordinate system [11]. Functioning as

a robust global prior, fingerprint pose plays a pivotal role in fingerprint recognition systems, and is typically employed as an essential preprocessing stage [11]. For example, pose normalization can substantially reduce intra-class differences caused by varying geometric positions, thereby effectively enhancing the generalizability and discriminability of feature extraction [8], [12], [13], [14], [15]. Besides, incorporating supplementary constraints on pose relationships inherently filters out imposter pairs with erroneous spatial correspondences while streamlining the search space, which can significantly improve the accuracy and efficiency of matching algorithms [10], [16], [17], [18], [19], [20].

Conventional approaches typically depend on special points [2], [4], [7] or specific areas [1], [5], [17], [21] to ascertain fingerprint pose. Motivated by the success of deep learning, researchers have gradually developed solutions based on neural networks in recent years [6], [8], [9], [10], [14], [20], [22]. However, these methods are initially designed for rolled or plain fingerprints, which generally necessitate a sufficiently large effective area (about 512×512 px, 500 ppi) and small angle differences (usually less than 30°) to acquire adequate information for reliable pose estimation. In addition to fingerprints, some studies proposed predicting the three-dimensional angle of fingers from capacitive images [23], [24], [25], which has been proven effective when inputted at relatively small touch angles (usually within 45°). It is imperative to highlight that the mobile device recognition scenario unequivocally surpasses these constraints. Specifically, with existing under-screen fingerprint sensors, the size of captured image is substantially reduced (about 132×132 px, 500 ppi), while users may attempt to unlock their devices from arbitrary touch poses, further exacerbating the challenges.

Fig. 1 shows examples of fingerprints in these mentioned modalities. To ensure terminological consistency, this paper adopts the following conventions: (1) *Plain Fingerprint*, *Ridge Patch* and *Capacitive Image* denote the specific image types used in distinct processing streams. (2) *Full Fingerprint* refers to a plain fingerprint that covers the complete contact area of a finger, distinguishing it from the smaller or lower-resolution samples captured by mobile devices. (3) *Partial Fingerprint* collectively describes the smartphone-based data, including both ridge patches and capacitive images. The correspondence among these terms is illustrated in Fig. 1. It can be intuitively seen that ridge patch and capacitive image exhibit a substantial degradation in terms of available information compared to plain fingerprint. Moreover, these two modalities from mobile devices exhibit significant differences: high resolution ridge

Received 5 May 2025; revised 30 October 2025; accepted 20 November 2025. Date of publication 25 November 2025; date of current version 5 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62376132 and Grant 62321005. The associate editor coordinating the review of this article and approving it for publication was Prof. Raghavendra Ramachandra. (*Corresponding author: Jianjiang Feng.*)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Institutional Review Board of Tsinghua University.

The authors are with the Department of Automation, Tsinghua University, Beijing 100084, China (e-mail: gxj21@mails.tsinghua.edu.cn; pzy20@mails.tsinghua.edu.cn; jfeng@tsinghua.edu.cn; jzhou@tsinghua.edu.cn).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TIFS.2025.3636671>, provided by the authors.

Digital Object Identifier 10.1109/TIFS.2025.3636671

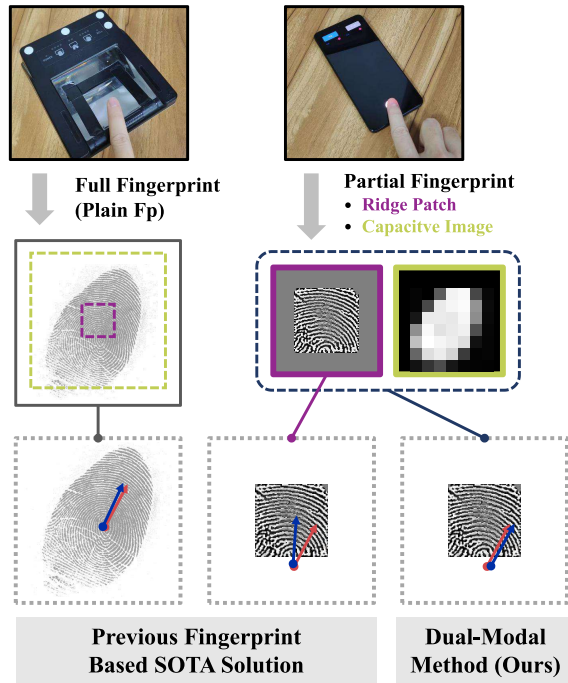


Fig. 1. Examples of fingerprint pose estimation under different input modalities. Among them, full fingerprint (plain fingerprint) is collected by a conventional optical fingerprint scanner, while the ridge patch and capacitive image are simultaneously collected from the under-screen fingerprint sensor and touch screen of a smartphone (referred to as partial fingerprint collectively). All modalities, captured from the same finger with similar touch gestures, are marked in gray, purple, and green. For clarity, the dashed lines in the full fingerprint indicate the equivalent collection areas for the partial fingerprints. Subfigures in the last row represent the estimated result (blue) and ground truth (red) using corresponding models. It can be observed that the performance of previous fingerprint-based SOTA solution [10] declines significantly as the available area diminishes, while our dual-modal method achieves more accurate prediction.

patches contain rich local structures but limited receptive field, while low resolution capacitive images roughly represent the global contour but lacking localized details. This exciting complementarity motivates us to explore the enormous potential of modal fusion implementation. What's more, the widespread adoption of mobile devices equipped with under-screen fingerprint sensors and touch screens has created an ideal ecosystem for applying this dual modal approach. These devices inherently support the simultaneous acquisition of ridge patches and capacitive images, making the proposed fusion method highly practical. Furthermore, this technology can be seamlessly integrated into existing devices through software updates, ensuring broad applicability across various scenarios. Overall, this innovative dual modal paradigm has convincing development value.

In this paper, we introduce a partial fingerprint pose estimation framework that effectively exploits such complementary strengths. The proposed approach leverages the collaborative potential of **Dual-modal** guidance from **Ridge** patches And **Capacitive** images to **Optimize** the feature extraction, fusion and representation, called **DRACO**. Different from the previous supervision forms of numerical regression [6], [8], [9], [14], [20] or heatmap voting [10], [22], we transform pose representations into decoupled quantized probability distri-

bution, inspired by [26], [27], [28], and [29]. This upgrade enables our network to better grasp the relative relationships between adjacent pose spaces, resulting in impressive performance gains. We also apply the MoE mechanism to improve the feature fusion stage. A lightweight router is employed to dynamically generate adaptive weights for multiple separated feature branches, ensuring an appropriate balance of significance across different modal information. Furthermore, we leverage the comparative relationships between groups to facilitate knowledge transfer from the high-performance plain fingerprint pose estimation domain to the target partial fingerprint domain, further strengthening the feature extraction part.

Extensive experiments are conducted on two public fingerprint databases and two private databases. The results strongly demonstrate the effectiveness of integrated strategies and mechanisms in DRACO. In addition, the proposed algorithm significantly outperforms existing SOTA pose estimation algorithms in terms of accuracy and stability. Moreover, we also evaluated the assistance of incorporating pose information for fingerprint recognition, where our approach demonstrates consistent leading performance.

The main contributions of this work can be summarized as:

- We propose DRACO, a dual-modal framework for partial fingerprint pose estimation with under-screen sensors. To the best of our knowledge, this is the first approach that jointly leverages ridge patch and capacitive image modalities, systematically exploiting their complementary geometric cues. By effectively integrating these two sources of information, DRACO achieves robust and accurate pose estimation under real-world under-screen conditions.
- Several mechanisms are introduced and adjusted to enhance feature extraction, fusion, and representation. Specifically, a Mixture of Experts (MoE) module enables adaptive modality fusion, while a relation-based knowledge transfer strategy guides the partial-fingerprint network to learn from the full-fingerprint domain, aligning relational structures to enhance feature representations. In addition, we propose a decoupled quantized probability representation that employs trigonometric encoding and Gaussian-weighted quantization for fine-grained and uncertainty-aware prediction.
- Extensive experiments are conducted on both simulated and real-world datasets, consistently surpassing existing state-of-the-art methods in terms of precision and robustness. Furthermore, our results demonstrate that accurate pose estimation can substantially enhance both full and partial fingerprint recognition, highlighting its great potential as an auxiliary task to improve recognition reliability and stability.

II. RELATED WORK

In this section, we first introduce the definition of fingerprint pose, and then review relevant finger pose estimation algorithms based on fingerprints (plain fingerprints or ridge patches) and capacitive images.

A. Definition of Fingerprint Pose

Owing to the absence of adequately distinct and consistent anatomical landmarks, the scientific community has yet to establish a clear and unified definition of fingerprint pose [11], [30]. Researchers have proposed multiple approaches to describe the center and direction of fingerprints in order to achieve approximate goals. Early studies employed the centroid and positive direction of foreground mask for pose estimation [3], [31], [32]. In addition, some approaches determine pose parameters through singular points [2], [33]. However, these approaches demonstrate unsatisfactory practicality, as their accuracy substantially depends on the completeness and quality of acquired fingerprint areas. To address these challenges, subsequent researchers proposed various fingerprint pose definitions based on special patterns along ridge orientation fields, such as points of maximum curvature [4], [34], points that match the reference templates [1], [31], or focal points perpendicular to the ridges [7], [35], [36]. Despite the improvement in accuracy, fingerprint centers under these definitions still lack sufficient consistency for different impressions.

Further integrating these features, Yang et al. [5] defined the fingerprint direction as perpendicular to the ridge orientation around the knuckle region, and determined the center based on the type and number of singular points. On this basis, Si et al. [37] introduced a refined approach that utilizes solely the central singular point located at the northernmost as the fingerprint center, while maintaining the same directional definition. In cases where such a singular point is absent, the point exhibiting the highest curvature is designated as the center. Subsequent studies [6], [9], [10], [17], [22] followed this form of definition. In the paper, we also adopt the same definition for consistency and comparability.

B. Pose Estimation Based on Fingerprint

Traditional methods typically estimate pose information through foreground mask information [3], [31], [32] or detecting special points [1], [2], [4], [7], [31], [33], [34], [35], [36]. However, such approaches demonstrate substantial deficiencies when confronted with incomplete or highly noisy data. Yang et al. [5] introduced a pose estimation algorithm utilizing Hough voting. During the offline phase, orientation fields are extracted from high-quality image patches to build region-specific dictionaries, which are then matched with input fingerprints during the online phase, with voting in Hough space and selecting the maximum response as the result. Similarly, Yin et al. [21] constructed a dictionary of global orientation fields from aligned high-quality fingerprints and made decisions through exhaustive search. Besides, Su et al. [17] employed Support Vector Machines (SVMs) to build a set of classifiers for identifying fingerprint center and direction using orientation field histograms. Furthermore, Gu et al. [22] utilized the orientation field and periodic map of ridge patches as features, and subsequently predicted the center position and direction based on the Hough Forest model and SVM, respectively. Despite tangible improvements, these region-based conventional machine learning approaches still

underutilize available data and achieve strong performance primarily on high-quality rolled fingerprints.

Over the past decade, deep learning based data-driven approaches have achieved impressive results across diverse domains. Ouyang et al. [6] decomposed the pose estimation task into object detection for position and classification for rotation, and introduced Faster-RCNN as the network structure. Schuch et al. [38] proposed an unsupervised learning paradigm, where a Siamese CNN is trained to predict the relative rotation between sample pairs and provide absolute angles during deployment. Yin et al. [9] proposed a multi-task network which simultaneously regresses the values of center, direction, and singular points of fingerprints. Furthermore, Arora et al. [39] suggested using two-stage prediction of core points through macro localization and micro-scale regression networks. Duan et al. [10] reformulated fingerprint pose estimation as dense prediction of grid offset vectors and employed a voting strategy. Moreover, some researchers developed several fingerprint descriptor extraction algorithms that include Spatial Transformer Networks (STN) [8], [14], [15], [40], where the byproducts of affine transformation parameters can be considered as a form of pose representation. These deep learning approaches have demonstrated remarkable success in processing both plain and rolled fingerprint images. Nevertheless, in the scenario of partial fingerprints, the substantial loss of available information presents a critical challenge that necessitates innovative approaches for effective resolution. Fig. 1 presents illustrative examples that offer qualitative validation of this perspective.

On the other hand, some studies proposed estimating relative pose from paired input fingerprints [20], [41], [42]. While the accuracy is notably improved, the trade-off involves additional relative alignment steps that must be executed separately for each comparison, significantly increasing the time cost of matching. Furthermore, these techniques necessitate the prior enrollment of sufficient fingerprint samples to facilitate the pose estimation of new impressions through comparative analysis. Therefore, this paper focuses exclusively on absolute pose estimation methods which are more efficient and less dependent, leaving the discussion and research of such schemes for future studies.

C. Pose Estimation Based on Capacitive Image

There are many studies on predicting the three-dimensional angle (yaw and pitch, without roll) of fingers during touch from capacitive images. Zaliva et al. [43] introduced multiple descriptive characteristics, such as area, centroid, and average intensity, to calculate finger angles. Xiao et al. [23] further defined 42 features and used Gaussian models to regress pitch and yaw angles. Subsequent studies [24], [25] used neural networks to directly predict finger angles from capacitive images, achieving the current SOTA performance. Due to the lack of discriminative texture information, these approaches are primarily applicable for small angle inputs (within $\pm 90^\circ$). In addition, inferring two-dimensional positions directly from low resolution contours (see Fig. 1) remains a major challenge.

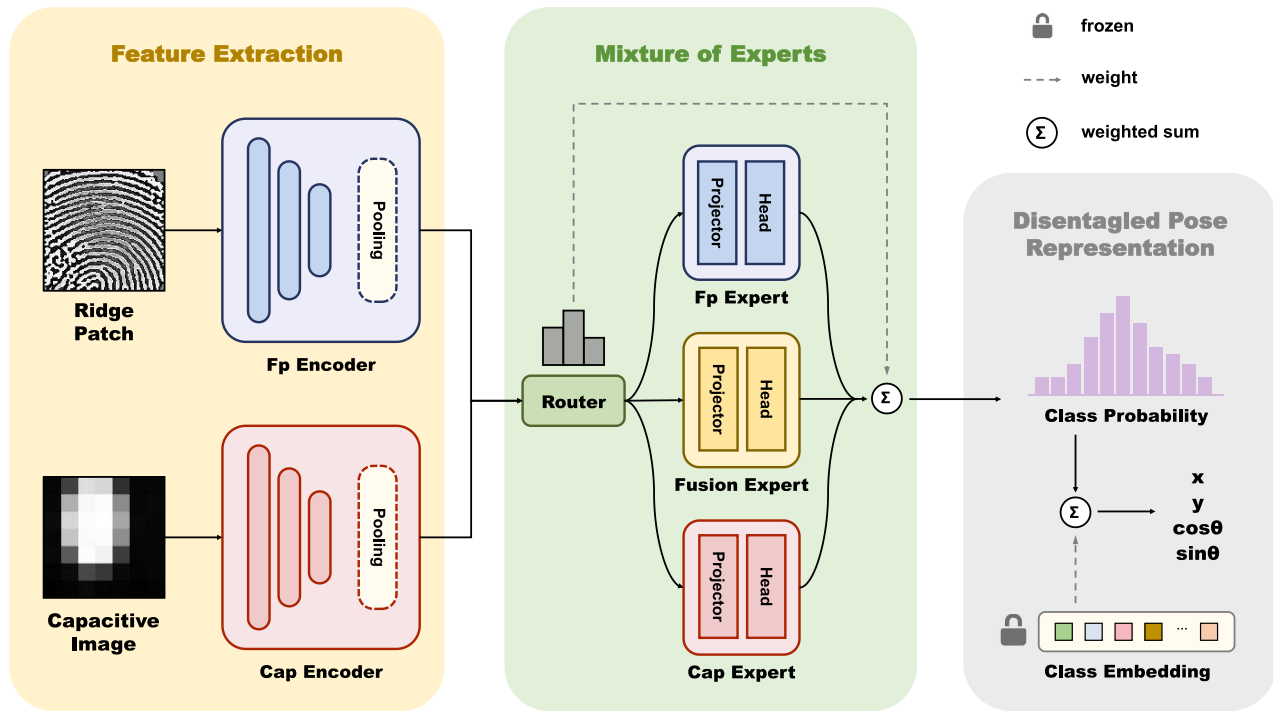


Fig. 2. An overview of our partial fingerprint pose estimation network DRACO. The ridge patch and capacitive image collected simultaneously from the touch device equipped with an under-screen fingerprint sensor are input. The prediction results are represented by the horizontal and vertical coordinates of the center, as well as the sine and cosine values of the direction.

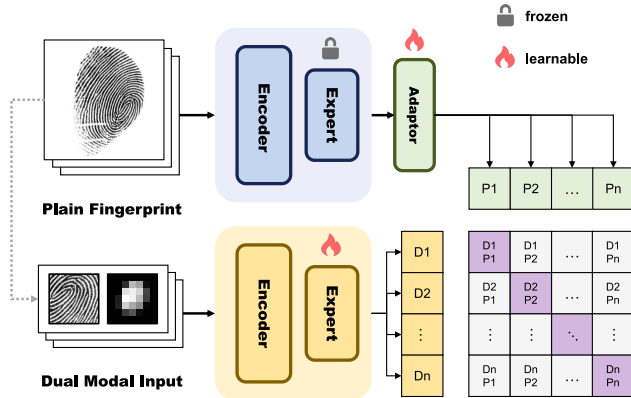


Fig. 3. The main process of knowledge transfer during network training. The blue and yellow modules correspond to modules of the same color in Fig. 2. As shown in the bottom right corner, the features of dual modal inputs are progressively aligned with the depiction in plain fingerprints through contrastive learning techniques. The gray dashed line represents the process of data synthesis.

III. METHOD

In this section, we will specifically introduce our partial fingerprint pose estimation method, which utilizes a dual modal input consisting of ridge patches and capacitive images. The proposed network, named DRACO, is shown in Fig. 2. Our framework employs a three-stage pipeline: (1) initially, two parallel branches are utilized to extract distinctive features from each modality; (2) subsequently, these features undergo integration through the MoE mechanism for comprehensive fusion and collaborative guidance; (3) ultimately, DRACO

generates decoupled pose representations in the form of independent one-dimensional probability distributions. Moreover, we design a knowledge transfer strategy from full fingerprints to partial fingerprints, as illustrated in Fig. 3, to help the network better comprehend and represent high-level semantic information as well as subtle differences between samples. These components will be sequentially introduced, and the corresponding loss functions are presented at the end. The construction of training data will be detailed in Section IV.

A. Feature Extraction

In this phase, two parallel encoders with homologous structures are utilized to separately derive modality-specific features from each input stream. Specifically, each branch begins with a stem that sequentially applies two consecutive groups of convolution, normalization, and activation operations. Given the success of ResNeXt-34 [44], we employ the same building blocks to construct a four-layer architecture with the configuration of [3, 4, 6, 3] [44], while incorporating spatial and channel attention modules [45] between layers to enhance the feature representation capabilities. It should be noted that when processing high-resolution ridge patches, the stride of each layer is set to 2 to facilitate downsampling [10], while remaining 1 for low-resolution capacitive images [25]. The extracted features are ultimately converted into corresponding one-dimensional vectors via global average pooling.

B. Mixture of Experts

Inspired by [46], [47], and [48], we introduced the MoE mechanism, which demonstrates significant efficacy in

addressing the inherent heterogeneity across different domains. The proposed architecture integrates three specialized experts: two dedicated to processing independent feature representations and one focused on mixed feature interactions, thereby flexibly enhancing the ability of multimodal feature fusion through collaborative guidance. A straightforward router, composed of two fully connected layers, dynamically generates adaptive weights for the inference of each expert. Let f^P and f^C represent the feature vectors of corresponding branches, the processing flow can be represented as:

$$\begin{aligned} f^F &= \text{cat}(f^P, f^C), \\ w^P, w^F, w^C &= \text{Router}[f^F], \end{aligned} \quad (1)$$

where $\text{cat}()$ corresponds to concatenation in the channel dimension. The subsequent feature enhancement and result assembly can be expressed as:

$$\begin{aligned} d^i &= \text{Head}^i[\text{Proj}^i[f^i]], \\ d &= \sum_{P,F,C} w^i \cdot d^i, \end{aligned} \quad (2)$$

where d is the corresponding pose parameter. To effectively capture the high-level semantic features while alleviating gradient-related issues, we stack one linear layer and four Multilayer Perceptrons (MLPs) with residual connections [49] as projector. On the other hand, a single linear layer is used as corresponding task head.

C. Disentangled Pose Representation

Departing from conventional numerical regression [6], [8], [9], [14], [20] or heatmap voting [10], [22] approaches, we reformulate pose parameters and supervision as decoupled probability distributions, thereby providing a more robust and interpretable representation. In other words, each sample yields four pose representations as output, corresponding to the horizontal and vertical coordinates of the fingerprint center, as well as the sine and cosine of the angle. Four sets of frozen category embeddings are pre-set to provide all spatial information. With this assistance, the model only needs to describe the similarity between pose information and each category, rather than directly predicting the specific encoding. This evolution can significantly alleviate the learning complexity and enhances the generalization capability [28], [29].

In this paper, we divide the horizontal and vertical displacements (from -256 px to 256 px) into 256 segments at equal intervals, and the sine and cosine (from -1 to 1) of the angle into 120 uniform segments as frozen class embeddings. These hyperparameters are primarily determined empirically through grid search on the training dataset. It is worth noting that trigonometric functions are used to represent fingerprint direction, instead of angle, to avoid confusion that may occur when approaching the two synonymous ends of 0° and 360° . Motivated by [26] and [27], we consider the class probability as a quantized distribution and perform weighted summation as the estimation result, rather than one-hot hard classification. For example, given the estimated probability distribution $\{d_t\}$

and class embedding $\{e_t\}$, the horizontal coordinate x of the fingerprint center can be calculated as

$$x = \frac{1}{\sum_{t=1}^n d_t} \sum_{t=1}^n d_t \cdot e_t, \quad (3)$$

where t and n correspond to the index and total number of segments, respectively. Taking this expectation helps avoid quantization errors caused by segment partitioning. The other pose parameters y , $\cos \theta$ and $\sin \theta$ are also obtained according to Equation 3. Additionally, the final direction θ is obtained by calculating the arctangent of these two trigonometric functions.

D. Knowledge Transfer

During the experiments, we observed that the same model demonstrated significantly superior performance on full fingerprint compared to partial fingerprint modalities. This phenomenon is reasonable because full fingerprints contain almost all the available one-sided features from both ridge patches and capacitive images, as well as richer and more comprehensive texture information. Therefore, we employ contrastive learning techniques [50], [51] to infuse global prior correlations (in plain fingerprint) into the current model applied to degraded modalities (partial fingerprint), leveraging a relationship-based knowledge transfer strategy [52], [53]. As shown in Fig. 3, a teacher model (blue) is pre-trained on plain fingerprints and its parameters are frozen. Next, a dual-modal student (yellow) is trained with the objective of extracting features whose relative relationships align as closely as possible with the judgments of the teacher. It should be noted that there is a strong correspondence between the dual modal input and plain fingerprint in each group, as the former is simulated from the latter and its process details are introduced in Section IV. In this paper, we use a network with the same structure as the fingerprint part in Fig. 2 as teacher, and reinforce the corresponding feature extraction and fusion experts through this task. Specifically, a stacked 2-layer MLP is used as adapter to appropriately adjust the inherent information gap between teacher and student. Under this relationship based supervision, features from different modals of the same impression are brought as close as possible, while features from distinct impressions are intentionally separated to maximize their distance. By leveraging these relational insights, the model gains a deeper understanding of data nuances, resulting in better outcomes without any additional cost during the testing stage.

E. Loss Function

For convenience, we jointly optimize the pose estimation and knowledge transfer process in one training process. As outlined in Section III-C, we employ the distance between quantitative probability distributions associated with each parameter as the supervisory signal for pose estimation task. Let Φ represent a certain pose component, d and $\tilde{d}$ represent the predicted result and ground truth of corresponding

TABLE I
ALL FINGERPRINT DATASETS USED IN EXPERIMENTS. AMONG THEM, PARTIAL FINGERPRINT REFERS TO TWO MODALITIES: RIDGE PATCHES AND CAPACITIVE IMAGES

Dataset	Type	Description	Usage	Genuine pairs ^a	Impostor pairs ^a
FVC2002 DB1_A [54]	Plain fingerprints with front pose	100 fingers $\times$ 8 impressions	test	n/a	n/a
FVC2004 DB1_A [55]	Plain fingerprints with front pose	100 fingers $\times$ 8 impressions	test	n/a	n/a
DPF [10]	Rolled fingerprints	933 fingers $\times$ 1 impression	calibration	n/a	n/a
	Plain fingerprints with diverse poses ^b	933 fingers $\times$ 3.1 impressions ^c	train & test	40,579	4,157,822
PCF	Rolled fingerprints	100 fingers $\times$ 1 impression	calibration	n/a	n/a
	Partial fingerprints with diverse poses	100 fingers $\times$ 32 impression	finetune & test	46,338	3,413,262

^a Effective pairs in matching experiments.

^b Simultaneously, partial fingerprints are simulated based on plain fingerprints.

^c Average number after screening.

probability distribution, the loss function of pose estimation $\mathcal{H}$ is defined as:

$$\mathcal{H} = \sum_{\Phi \in \{x, y, \cos, \sin\}} \lambda_{\Phi} \cdot \text{dist}(d_{\Phi}, \tilde{d}_{\Phi}), \quad (4)$$

where Cross Entropy (CE) serves as the distance metric $\text{dist}(\cdot)$ between two distributions. All balance factors λ are empirically set to 1.0. The value $\tilde{v}$ of ground truth is converted into discrete probability $\tilde{d}$ using gaussian distribution:

$$\tilde{d}_t = \exp\left(-\frac{\tilde{v} - e_t}{2\sigma^2}\right) / \sum_i \tilde{d}_t, \quad (5)$$

where the definition of e_t is consistent with Equation 3, and the hyperparameter σ is set to 3.5 and 2.5 for position and angle sub-losses. To further augment the individual capabilities of each expert, we integrate classification heads with the same structure after each expert branch as subtasks. The total loss of the pose estimation part is

$$\mathcal{L}_{\text{pose}} = \sum \lambda_e \cdot \mathcal{H}_e, \quad (6)$$

where the hyperparameters λ_e corresponding to the three experts (Fp, Fusion and Cap in Fig. 2) and the comprehensive results, fixed as 0.2, 0.2, 0.4, and 1.0, respectively. It is worth mentioning that above hyperparameters are not specially tuned but rather follow standard practice to maintain comparable magnitudes among loss terms. This design ensures balanced optimization without introducing dataset-specific bias.

On the other hand, the Information Noise Contrastive Estimation (InfoNCE) with temperature coefficients is used to optimize the knowledge distillation process in Section III-D. Referring to Fig. 3, a plain fingerprint and the corresponding simulated dual modal images are used as the input group during training. For the features $\{P\}$ and $\{D\}$ extracted by the teacher and student networks, this relationship-based supervision is computed as:

$$z(D_i, P) = \frac{\exp(\text{sim}(D_i, D_i^+) / \tau)}{\sum_{j=1}^B \exp(\text{sim}(D_i, P_j) / \tau)},$$

$$\mathcal{L}_{\text{KT}} = -\frac{1}{2B} \sum_i (\log z(D_i, P) + \log z(P_i, D)), \quad (7)$$

where B is the batch size, sim is the cosine similarity. The temperature τ is set to 8.0 according to the results of small-scale parameter search.

The final loss is computed as the weighted sum of aforementioned two tasks:

$$\mathcal{L} = \mathcal{L}_{\text{pose}} + \lambda \cdot \mathcal{L}_{\text{KT}}. \quad (8)$$

The balance term λ is configured to 1.0.

IV. DATASET

A total of four datasets are used in our experiments, including two public datasets (*FVC2002 DB1_A* [54] and *FVC2004 DB1_A* [55]), one non-public dataset (*DPF* [10]), and one collected by ourselves (*PCF*). The utilization of all datasets is summarized in Table I, and Fig. 4 presents representative image examples. In the following subsections, we first describe the key characteristics of these datasets and then detail the construction of training and testing sets.

A. Dataset Introduction

1) *FVC2002 DB1_A* & *FVC2004 DB1_A*: Since most existing public datasets are predominantly collected under frontal poses and lack diversity in acquisition conditions, we only selected two widely used datasets—*FVC2002 DB1_A* [54] and *FVC2004 DB1_A* [55]—as representatives. Most fingerprints in these databases are captured under frontal poses, with multiple impressions recorded for each finger. Following previous studies [6], [8], [9], [10], we employ these datasets to evaluate the performance of pose estimation algorithms in relatively simplified scenarios, i.e., full fingerprint pose estimation.

2) *DPF*: Unlike the above datasets, the private *DPF* database [10] was specifically collected with subjects instructed to adopt diverse finger poses, making it more suitable for pose estimation tasks. In our experiments, 8,479 samples from 744 fingers were used for training, while the remaining 2,049 samples from another 189 fingers were reserved for testing. The performance of the pose estimation algorithms was evaluated on these full-fingerprint data, and additional experiments were conducted on simulated partial-fingerprint cases.

3) *PCF*: In addition, we collected and established a real partial fingerprint dataset using a smartphone equipped with an under-screen fingerprint sensor. This dataset, referred to as the Phone Captured Fingerprint database with matching

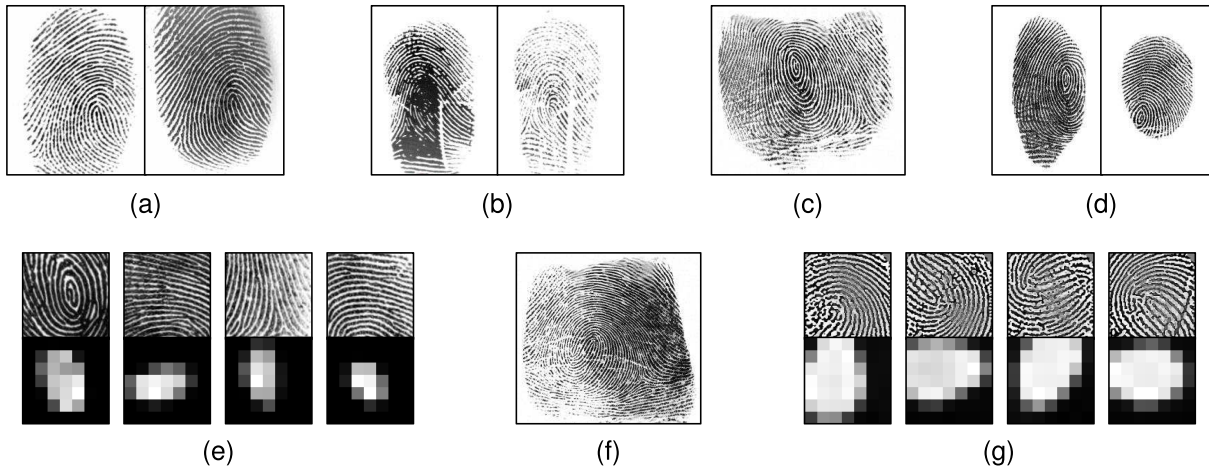


Fig. 4. Image examples from different datasets: (a) FVC2002 DB1_A [54], (b) FVC2004 DB1_A [55], (c) DPF [10], rolled fingerprint, (d) DPF [10], plain fingerprints, (e) DPF [10], simulated partial fingerprints, (f) PCF, rolled fingerprints, (g) PCF, partial fingerprints. The ‘partial fingerprint’ is a general term used to represent two modalities: ridge patch and capacitive image.

rolled fingerprint (*PCF*), enables the simultaneous acquisition of ridge patches and capacitive images. The details of collection equipment and process are provided in Appendix A of the Supplementary Materials. Specifically, ridge patches of 132×132 pixels at 500 ppi and capacitive images of 7×7 pixels (effective area) at 10 ppi were captured concurrently. A total of 640 images from 20 fingers were used for model fine-tuning, while 2,560 images from another 80 fingers were reserved for testing. This dataset was employed to evaluate the performance of pose estimation algorithms under challenging real-world partial fingerprint scenarios.

In both the *DPF* and *PCF* datasets, the subjects were also asked to provide rolled fingerprints, which were used for subsequent calibration to obtain pose ground truth. All participants provided informed consent for the use of their data in this research.

B. Training Set Construction

We use the subset of *DPF* mentioned above for training. For full fingerprint scenarios, we directly utilize the samples displayed in Fig. 4(d) for training. The ground truth is obtained by minutiae matching between the input plain fingerprint and its corresponding pose-standardized rolled fingerprint. In this paper, we employ VeriFinger SDK 12.0 [56] for minutiae extraction and matching, and adopt the method proposed by Duan et al. [10] for rolled fingerprint pose rectification. These approaches are proven to be sufficiently reliable under conditions of high image quality and small pose variation [10]. Therefore, we approximately consider it as an unbiased and precise calibration.

Due to the scarcity of readily available large-scale partial fingerprint data and capacitive images, we generated approximate samples from the plain fingerprints of *DPF*. Following [41], [42], and [57], we randomly crop square regions from the plain fingerprint to simulate ridge patches. On the other hand, window-based uniform filtering and interpolation are applied to downsample the original image to 10 ppi, emulating the capacitive image [57]. The setting of cropping size and target

resolution is to maintain consistency with the real data in *PCF*. Nevertheless, we fine-tuned our model using a small amount of local data before testing on *PCF* to minimize domain bias as much as possible.

C. Test Set Protocol

To comprehensively evaluate the accuracy and robustness of pose estimation algorithms, we generated four distinct test sets for each scenario. Specifically, the pose of samples in *FVC2002 & FVC2004 DB1_A* and *DPF* is first standardized and then randomly rotated within the four ranges of $\pm 45^\circ$, $\pm 90^\circ$, $\pm 135^\circ$ and $\pm 180^\circ$. Each evaluated method will be trained and tested separately based on the range of direction angle under full fingerprint or partial fingerprint scenario. For simplicity, the training conditions of each model will not be specifically declared in experiments. Additionally, the real partial fingerprint data *PCF* remains unchanged in order to accurately provide feedback on the real environment. The ground truth of its pose information is obtained through the same calibration process introduced in Section IV-B. In the matching experiments, pairs with the same identity will be checked in advance to exclude situations where there is no overlapping area. The effective number of genuine and impostor pairs is presented in Table I.

V. EXPERIMENTS

In experiments, we compare the proposed DRACO with SOTA finger pose estimation algorithms on full fingerprints (plain fingerprints) and partial fingerprints (ridge patches and capacitive images). In the following, we first present the implementation details of DRACO, followed by a description of the representative methods used for comparison. The performance of these algorithms is then thoroughly evaluated in terms of pose estimation and matching capabilities across various rotation ranges. Extensive ablation studies are also conducted to validate the effectiveness of the proposed modules and strategies, while offering insights for future research. Finally,

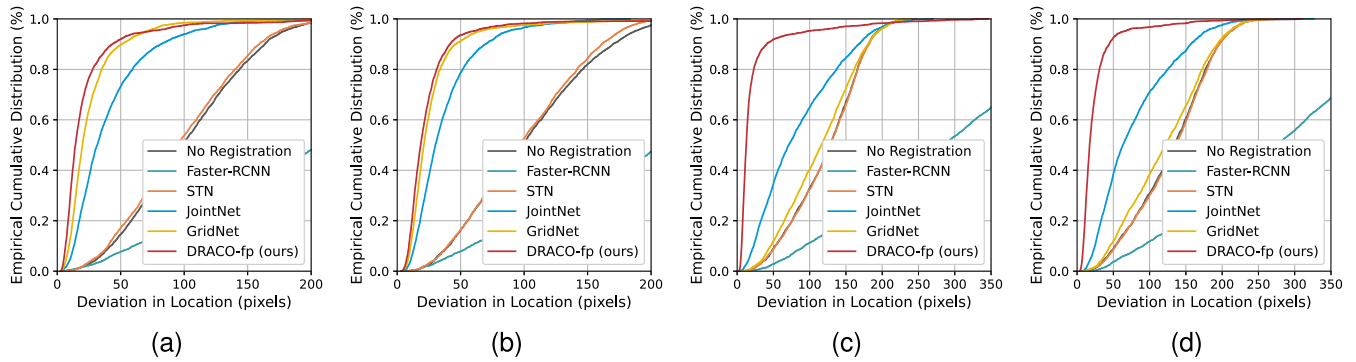


Fig. 5. The empirical cumulative distribution of location deviations on full fingerprints from (a) FVC2002 DB1_A $[-90^\circ, 90^\circ]$, (b) FVC2004 DB1_A $[-90^\circ, 90^\circ]$, (c) FVC2002 DB1_A $[-180^\circ, 180^\circ]$, (d) FVC2004 DB1_A $[-180^\circ, 180^\circ]$. Suffixes ‘fp’ indicate that only the corresponding branch is used.

the efficiency of the different algorithms is reported to assess the potential deployment costs.

A. Implementation Details

Our proposed DRACO is trained under the corresponding modality in *DPF* with an initial learning rate of $1e-3$ (end of $1e-6$), cosine annealing scheduler, default AdamW optimizer and batch size of 128 for 80 epochs. Data augmentation is used, including random translation within ± 40 pixels. Additionally, the input is randomly rotated within $\pm 45^\circ$, $\pm 90^\circ$, $\pm 135^\circ$, and $\pm 180^\circ$, according to the corresponding rotation range scenario. Specifically, when incorporating contrastive learning, we increase the batch size to 512 to ensure sample richness. The learning rate and epoch number are adjusted to $4\times$ and 200 to roughly maintain the original optimization iterations. Before testing on *PCF*, the parameters under $\pm 180^\circ$ augmentation are loaded and further fine-tuned with an initial learning rate of $1e-4$ (end of $1e-5$) for 200 epochs, while keeping other parameters consistent. When DRACO is applied in a single modal, only the corresponding feature extractor and expert shown in Fig. 2 are activated. The training hyperparameters are set following standard deep learning practices, without any special tuning. All hyperparameters are fixed globally across datasets and models to ensure fairness and consistency in the experimental comparisons. In following experiments, we used suffixes ‘fp’ and ‘cap’ to distinguish models with DRACO structures that only use branch of ridge patch and capacitive image, respectively.

B. Compared Methods

For plain fingerprint and ridge patch, we reproduced four representative methods, including:

- **Faster-RCNN**: Object detection network proposed by Ouyang et al. [6];
- **STN**: STN module under indirect supervision of fixed-length representation task [8], [14];
- **JointNet**: Numerical regression network proposed by Yin et al. [9];
- **GridNet**: Heatmap voting network recently proposed by Duan et al. [10].

Additionally, for the capacitive image modality, since existing researches focus on predicting angles in 3D space, we

re-implement the following approaches and adjust the output head to estimate the center and rotation value of 2D pose:

- **Cap-MLP**: Regressor based on manually defined features and multi-layer MLP, inspired by Xiao [23];
- **Cap-CNN**: Numerical regression network inspired by recent works [24], [25].

Among them, we obtained the official training and model codes of [9], [10], and [25]. For other methods that are not open source [6], [8], [14], [23], [24], we reimplemented them according to the descriptions in their respective papers. To ensure fair and consistent comparisons, all baseline methods were retrained using the same dataset configuration, preprocessing pipeline, and hyperparameter settings as our proposed method.

C. Pose Estimation Performance

1) *Evaluation Protocols*: Although the focus of this paper is on partial fingerprint pose estimation, our disentangled pose representation strategy is universal. Therefore, we first compare representative methods with our proposed DRACO in full fingerprint scenario, and then provide detailed evaluation on partial fingerprint (including partial fingerprint and capacitive image) pose estimation. Due to the absence of ground-truth pose annotations in full fingerprint datasets, we evaluate the estimated pose consistency by measuring the deviations in direction and location between different samples of the same finger. Empirical cumulative distribution plots are then drawn for these deviations, following the evaluation protocol used in previous studies [9], [10]. On the other hand, in the partial fingerprint scenario, the ground-truth pose values are available through the scheme introduced in Section IV-A. Therefore, the average errors of N samples can be directly computed as:

$$e_{\text{trans}} = \frac{1}{N} \sum_i \|\tilde{p}_i - p_i\|, \quad e_{\text{rot}} = \frac{1}{N} \sum_i \min(|\tilde{\theta}_i - \theta_i|, 360 - |\tilde{\theta}_i - \theta_i|), \quad (9)$$

where subscripts trans and rot denote translation and rotation deviations, p and θ are the position and direction of the fingerprint pose, and the superscript, $\sim$, indicates the predicted value; otherwise, it denotes the ground truth.

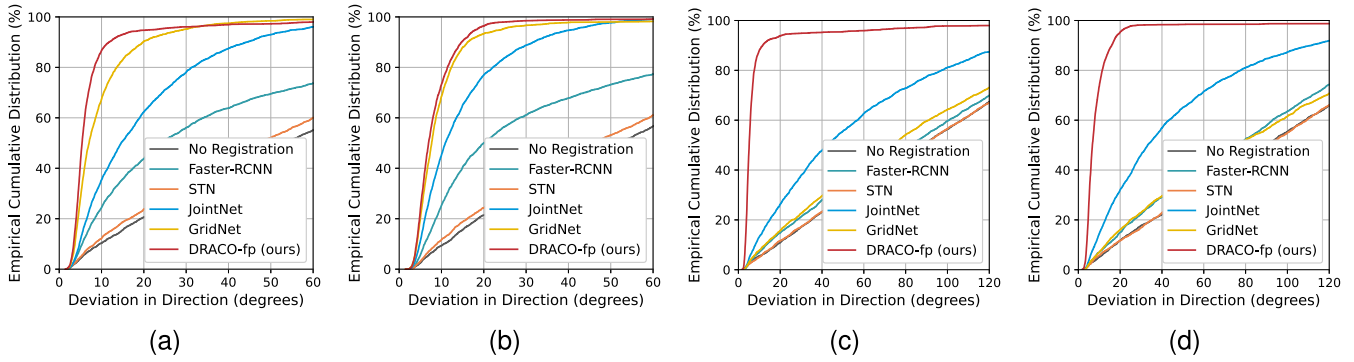


Fig. 6. The empirical cumulative distribution of direction deviations on full fingerprints from (a) FVC2002 DB1_A $[-90^\circ, 90^\circ]$, (b) FVC2004 DB1_A $[-90^\circ, 90^\circ]$, (c) FVC2002 DB1_A $[-180^\circ, 180^\circ]$, (d) FVC2004 DB1_A $[-180^\circ, 180^\circ]$. Suffixes ‘fp’ indicate that only the corresponding branch is used.

TABLE II
ALIGNMENT ERROR UNDER DIFFERENT FINGERPRINT POSES ON DPF (FULL FINGERPRINTS WITH DIFFERENT ROTATION RANGES).
SUFFIXES ‘fp’ INDICATE THAT ONLY THE CORRESPONDING BRANCH IS USED

Method	$[-45^\circ, 45^\circ]$		$[-90^\circ, 90^\circ]$		$[-135^\circ, 135^\circ]$		$[-180^\circ, 180^\circ]$	
	trans (px)	rot ($^\circ$)	trans (px)	rot ($^\circ$)	trans (px)	rot ($^\circ$)	trans (px)	rot ($^\circ$)
No Registration	83.4	22.7	85.5	45.5	85.1	67.4	88.3	90.8
Faster-RCNN [6]	218.4	22.3	224.4	27.0	118.3	66.6	221.5	100.1
STN [8], [14]	90.1	16.2	85.9	41.8	85.7	67.6	89.2	92.0
JointNet [9]	16.3	4.6	18.8	7.5	23.0	13.5	22.8	19.9
GridNet [10]	14.9	4.2	19.7	6.2	70.9	28.6	83.7	62.3
DRACO-fp (ours)	14.5	1.8	15.0	2.0	14.1	1.8	14.2	1.8

2) *Evaluation on Full Fingerprints*: Consistent with previous works [9], [10], the deviation in location and direction of mated minutiae pairs on *FVC2002 DB1_A* and *FVC2004 DB1_A* is presented to characterize the accuracy of pose estimation. Specifically, each method infers the 2D pose of single plain fingerprint, which is then utilized to execute a rigid transformation and align the image to standard coordinate system. Subsequently, VeriFinger [56] is used to extract and match minutiae between impressions of the same identity. The average Euclidean distance deviation and absolute rotation error between paired points is visualized through empirical cumulative score function. As shown in Fig. 5 and Fig. 6, the proposed DRACO exhibits significant advantages across all test datasets, especially when dealing with large rotation angles. The performance of *STN* [8], [14] is close to no registration, primarily because pose estimation is not directly supervised. This absence of explicit guidance may impede the network’s ability to learn accurate transformations. In addition, the object detection based *Faster-RCNN* [6] shows limited performance in estimating fingerprint position. A compelling explanation is that in this task, the network primarily focuses on the outer contours, which may lead to neglect of the precise center.

Furthermore, we evaluated the performance of these algorithms across four different rotation ranges on plain fingerprints of *DPF*. The mean absolute error of translation and rotation is reported in Table II. It can be seen that the performance of previous SOTA methods significantly declines as the rotation range increases. In contrast, our method demonstrates exceptional precision and robustness

across the entire rotation range. Based on these experimental results, we selected the best-performing algorithms, *JointNet* [9] and *GridNet* [10], for the subsequent experiments on partial fingerprints.

3) *Evaluation on Partial Fingerprints*: Similarly, we assessed the performance of pose estimation algorithms on partial fingerprints (partial fingerprints and capacitive images) using the simulated test set from *DPF*, following the protocol defined in Equation 9. As demonstrated in Table III, our proposed solution outperforms previous methods in respective unimodal groups. This result strongly emphasizes the advantages of our pose representation scheme once again. In addition, the full version of DRACO showcases comprehensive and notable superiority in both accuracy and stability following the integration of dual modal information. This substantial improvement clearly demonstrates the considerable complementarity between ridge patches and capacitive images, affirming the effectiveness of the collaborative dual-modal guidance paradigm. Table IV presents the key comparative results on *PCF*, which also highlight the impressive success of our proposed algorithm. The overall performance on *PCF* is somewhat inferior to the simulation dataset of *DPF*, possibly due to domain bias and the limited fine-tuning data (640 samples during fine-tuning, which is significantly smaller than the 8,479 samples in training stage). However, this experiment still provide valuable insights into evaluating the relative performance of pose estimation algorithms, which is our primary concern.

4) *Visual Analysis*: Several intuitive examples are provided to qualitatively compare different pose estimation algorithms.

TABLE III

ALIGNMENT ERROR UNDER DIFFERENT FINGERPRINT POSES ON DPF (PARTIAL FINGERPRINTS WITH DIFFERENT ROTATION RANGES). THE FOUR GROUPS, FROM TOP TO BOTTOM, REPRESENT DEFAULT CONFIGURATION AND METHODS THAT USE ONLY RIDGE PATCHES, ONLY CAPACITIVE IMAGES, AND A COMBINATION OF BOTH. SUFFIXES 'fp' AND 'cap' INDICATE THAT ONLY THE CORRESPONDING BRANCH IS USED. BOLD AND UNDERLINED NUMBER REPRESENT THE CORRESPONDING GLOBAL OPTIMAL RESULT AND OPTIMAL METHOD IN EACH GROUP RESPECTIVELY

Method	[−45°, 45°]		[−90°, 90°]		[−135°, 135°]		[−180°, 180°]	
	trans (px)	rot (°)	trans (px)	rot (°)	trans (px)	rot (°)	trans (px)	rot (°)
No Registration	83.4	22.7	85.5	45.5	85.1	67.4	88.3	90.8
JointNet [9]	34.3	17.3	36.1	18.7	37.3	27.9	41.4	34.5
GridNet [10]	35.2	12.0	44.2	19.6	47.3	37.8	74.8	64.5
DRACO-fp (ours)	23.4	10.3	22.6	11.7	26.8	15.1	25.4	14.1
Cap-MLP [23]	79.7	13.3	79.3	39.2	83.5	65.5	86.2	91.1
Cap-CNN [24], [25]	67.8	7.8	<u>66.2</u>	16.0	71.7	58.5	78.3	70.4
DRACO-cap (ours)	<u>65.0</u>	<u>7.1</u>	67.8	<u>12.9</u>	<u>70.8</u>	<u>49.7</u>	<u>75.7</u>	<u>68.1</u>
DRACO (ours)	18.4	4.8	19.5	5.4	20.3	5.5	19.8	5.5

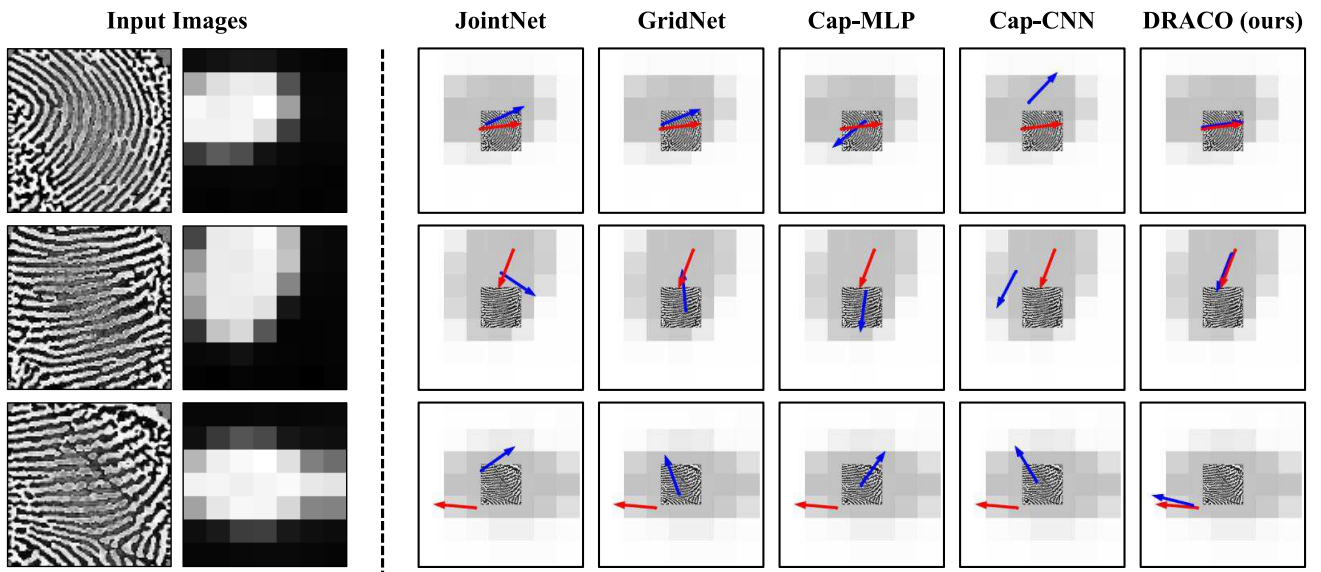


Fig. 7. Examples of different pose estimation methods on partial fingerprints from the *PCF* dataset. The two leftmost columns show the ridge patch and capacitive image as inputs. Each subsequent column on the right corresponds to a specific method (JointNet [9], GridNet [10], Cap-MLP [23], Cap-CNN [24], [25], and the proposed method). For visualization, all methods share the same background, where the input ridge patch is overlaid with inverted capacitive image. The predicted poses and ground-truth are indicated by blue and red arrows, respectively.

TABLE IV

ALIGNMENT ERROR UNDER DIFFERENT FINGERPRINT POSES ON PARTIAL FINGERPRINTS FROM PCF. THE GROUPING RULES ARE THE SAME AS IN TABLE III

Method	trans (px)	rot (°)
No Registration	98.4	82.0
JointNet [9]	51.5	38.0
GridNet [10]	87.2	63.2
Cap-MLP [23]	94.8	75.9
Cap-CNN [24], [25]	90.6	68.2
DRACO (ours)	31.2	16.3

Fig. 7 shows three representative visualization results. It can be observed that fingerprint based methods (JointNet [9], GridNet [10]) function effectively when the texture features of ridge patches possess sufficient discriminability (line 1). On the other hand, it is possible to accurately infer

angles using only capacitive images (Cap-MLP [23], Cap-CNN [24], [25]), which performs significantly better than relying solely on ridge patches (line 2). However, capacitive image based methods expose obvious deficiencies in localization. The respective advantages and limitations of these two modals effectively highlight their complementarity. Naturally, our method, guided by dual-modal collaboration, demonstrates more precise performance (line 1 & 2). Even when previous solutions completely fail, it can still provide accurate predictions (line 3).

We further illustrate some failure cases of DRACO in Fig. 8. Under certain extreme pressing postures, such as those involving fingertips (columns 1 & 2), the model may encounter substantial pose estimation errors. Additionally, in rare instances where both ridge patches and capacitive images lack sufficient recognition, DRACO may occasionally experience considerable confusion and misjudgments. (column 3).

TABLE V

VERIFICATION PERFORMANCE (%) ON DPF (FULL FINGERPRINTS WITH DIFFERENT ROTATION RANGES) USING DIFFERENT FINGERPRINT POSES. SUFFIXES 'FP' INDICATE THAT ONLY THE CORRESPONDING BRANCH IS USED

Matcher	Method	[-45°, 45°]			[-90°, 90°]			[-135°, 135°]			[-180°, 180°]		
		EER	FNMR ¹	FNMR ²	EER	FNMR ¹	FNMR ²	EER	FNMR ¹	FNMR ²	EER	FNMR ¹	FNMR ²
VeriFinger [56]	No Registration	0.71	0.42	1.90	0.71	0.55	1.95	0.64	0.34	1.84	0.66	0.43	1.84
	JointNet [9]	0.64	0.32	1.90	0.64	0.32	1.95	0.60	0.23	1.66	0.85	0.79	2.20
	GridNet [10]	0.63	0.32	1.90	0.63	0.32	1.95	0.70	0.48	1.98	4.07	4.43	5.81
	DRACO-fp (ours)	0.59	0.26	1.82	0.61	0.24	1.76	0.54	0.15	1.66	0.54	0.13	1.70
FDD [15]	No Registration	30.66	61.02	69.73	40.14	83.44	89.03	44.41	90.05	94.45	44.95	92.80	96.40
	JointNet [9]	1.53	1.72	4.34	2.67	3.79	7.06	6.72	11.66	19.05	12.19	22.13	30.36
	GridNet [10]	1.34	1.49	3.62	2.57	3.48	5.84	30.00	55.44	63.11	41.40	85.19	90.73
	DRACO-fp (ours)	1.32	1.50	3.48	1.19	1.35	3.67	1.18	1.29	3.65	1.07	1.14	3.34

¹ FNMR@FMR=1e-3, ² FNMR@FMR=1e-4.

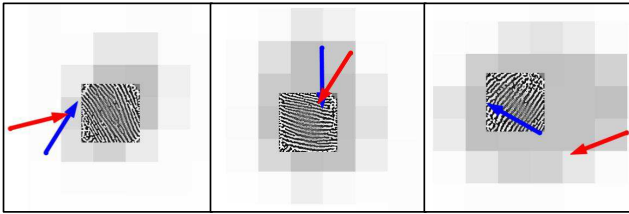


Fig. 8. Failure cases of DRACO on partial fingerprints from PCF. The visualization protocol is consistent with Fig. 7.

D. Matching Performance

1) *Evaluation Protocols*: Fingerprint pose estimation serves as a valuable source of auxiliary information in matching tasks [10], [11], [17]. For instance, an impression taken from the left side of a finger should not be considered a successful match with any impression from the right side, no matter how similar they are. According to this logic, recognition systems can swiftly identify and discard candidate samples that exhibit incompatible positions and orientations. This selective filtering significantly reduces the search space, leading to enhanced matching accuracy and efficiency in the overall process.

In experiments, we use VeriFinger [56] as a representative keypoint based matcher, which provides high quality minutiae extraction and matching functions. Samples with pose differences greater than an optimal threshold (determined through exhaustive parameter search) are excluded in advance. In other words, the comparison scores of these detected abnormal situations are set to infinitely small. On the other hand, we also assessed the impact of pose rectification on recognition schemes using fixed-length representations. The current SOTA method FDD [15] is utilized on behalf of these approaches. In the recognition process, the input image is first rigidly transformed based on corresponding estimated pose. Subsequently, one-dimensional representation vectors are extracted and the matching similarity between pairs is calculated.

Following previous studies [9], [10], [11], we adopt EER (Equal Error Rate), and FNMR @ FMR (False Non-Match Rate at specific False Match Rate) as evaluation metrics. In addition, indexing curves are provided to more comprehensively evaluate and compare the performance of different methods.

2) *Evaluation on Fingerprint Verification*: Consistent with Section V-C, we evaluate the performance of pose estimation methods on both plain and partial fingerprints. This enables us to thoroughly evaluate their effectiveness across different situations. The results across different rotation ranges on DPF are reported in Table V and VI, respectively. Additionally, Table VII presents a further comparison on PCF. When the pose difference is small, GridNet [10] exhibits certain advantages. As the rotation angle increases, the performance of JointNet [9] becomes more stable. In more challenging scenarios involving partial fingerprints, previous works, whether based on ridge patches or capacitive images, have demonstrated unsatisfactory performance. At the same time, our proposed DRACO showcases impressive and comprehensive advancements. We attribute it to the decoupled pose representation and collaborative dual-modal guidance developed in this paper. It is worth mentioning that fixed-length representation based matcher [15] still has a gap in partial fingerprints compared to minutiae based solutions [56]. Nevertheless, the introduction of our pose estimation method shows a significant relative improvement and attractive potential for future refinements.

3) *Evaluation on Fingerprint Indexing*: Similarly, we further examine the role of fingerprint pose estimation algorithms within the indexing system. Experiments are conducted solely on the minutiae based matcher, as this approach demonstrates superior performance and is the most commonly used in related works [11]. Fig. 9 and 10 illustrate the indexing performance for full and partial fingerprint scenarios on DPF, respectively, across different rotation ranges. The results on PCF are shown in Fig. 11. Comparisons in these curves indicate that appropriate pose estimation can effectively improve the indexing accuracy. In scenarios with a large rotation range or restricted effective areas, previous works face greater challenges and even exhibit negative impacts in some cases. Notably, our method excels as the most stable and accurate across all scenarios, which shows consistent positive effects in all tests.

E. Ablation Study

1) *Evaluation Protocols*: In this subsection, we conduct ablation experiments separately on the pose representation

TABLE VI

VERIFICATION PERFORMANCE (%) ON DPF (PARTIAL FINGERPRINTS WITH DIFFERENT ROTATION RANGES) USING DIFFERENT FINGERPRINT POSES. THE FOUR GROUPS, FROM TOP TO BOTTOM, REPRESENT DEFAULT CONFIGURATION AND METHODS THAT USE ONLY RIDGE PATCHES, ONLY CAPACITIVE IMAGES, AND A COMBINATION OF BOTH

Matcher	Method	[−45°, 45°]			[−90°, 90°]			[−135°, 135°]			[−180°, 180°]		
		EER	FNMR ¹	FNMR ²	EER	FNMR ¹	FNMR ²	EER	FNMR ¹	FNMR ²	EER	FNMR ¹	FNMR ²
VeriFinger [56]	No Registration	2.24	4.39	13.94	2.06	3.88	14.81	2.24	4.74	15.14	2.32	5.19	17.17
	JointNet [9]	2.13	3.84	13.08	2.07	3.70	14.19	2.60	4.81	15.74	2.87	5.71	18.08
	GridNet [10]	2.06	3.42	13.12	2.02	3.59	14.10	2.78	5.23	16.08	3.71	7.01	19.19
	Cap-MLP [23]	2.24	3.80	13.44	2.35	4.50	15.37	6.56	10.82	20.68	6.31	10.21	20.93
	Cap-CNN [24], [25]	2.04	3.40	13.11	2.12	4.05	14.97	4.80	8.64	18.74	4.56	8.04	20.16
	DRACO (ours)	1.27	1.61	9.50	1.11	1.30	8.41	1.13	1.26	8.92	0.96	0.92	9.09
FDD [15]	No Registration	25.43	67.77	85.32	36.67	84.20	93.53	44.30	90.29	96.03	44.05	92.49	97.46
	JointNet [9]	20.72	53.97	74.40	17.42	42.98	62.33	23.74	56.50	74.82	28.90	67.27	83.11
	GridNet [10]	14.34	38.54	58.24	19.04	47.84	66.17	31.50	69.64	82.82	40.46	85.99	93.74
	Cap-MLP [23]	23.82	58.48	76.88	36.05	82.12	91.37	44.62	88.56	95.10	44.15	90.60	96.58
	Cap-CNN [24], [25]	23.66	59.14	78.27	27.12	61.97	79.13	43.33	88.71	95.31	44.14	92.18	97.34
	DRACO (ours)	5.70	16.71	37.41	6.40	19.48	40.47	6.75	20.39	44.14	6.87	21.37	46.42

¹ FNMR@FMR=1e-3, ² FNMR@FMR=1e-4.

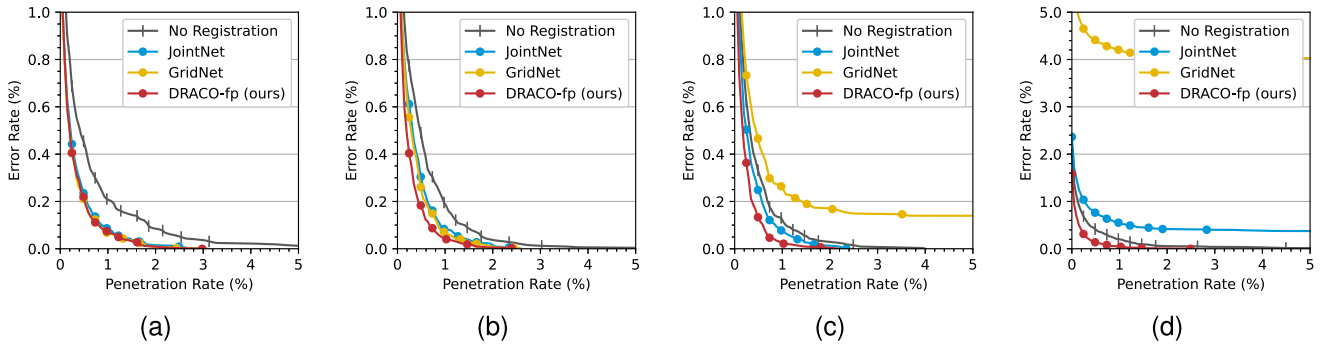


Fig. 9. The VeriFinger [56] based fingerprint indexing performance with corresponding pose constraint on full fingerprints from *DPF* under different rotation ranges: (a) [45°, 45°], (b) [90°, 90°], (c) [135°, 135°], (d) [180°, 180°]. Different input modalities are distinguished by the shape of markers.

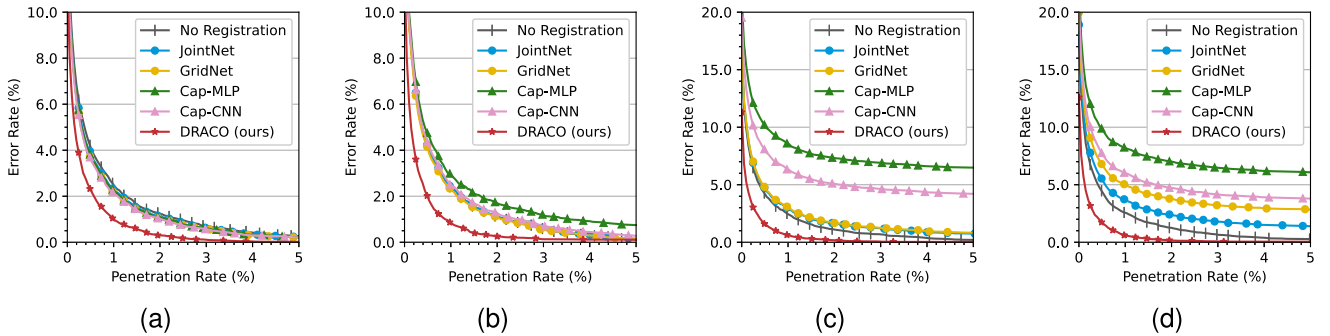


Fig. 10. The VeriFinger [56] based fingerprint indexing performance with corresponding pose constraint on partial fingerprints from *DPF* under different rotation ranges: (a) [−45°, 45°], (b) [−90°, 90°], (c) [−135°, 135°], (d) [−180°, 180°]. Different input modalities are distinguished by the shape of markers.

form and the network architecture design. Specifically, according to Equation 9, the average angular estimation error is used as a representative metric to compare the effects of different representation forms on prediction accuracy. In addition, based on the same equation, we evaluate the influence of different strategies and architectures on both translation and rotation errors.

2) *Pose Representation Form*: We first investigate the impact of pose representation and supervision forms on plain

fingerprints of *DPF*. Table VIII presents the ablation study for rotation as a representative example. For the regression task head, two specific expressions are compared: (1) *ang*: directly predicting the angle, and (2) *tan*: predicting sine and cosine components, with the angle computed via the arctangent function. Similarly, two types of mean squared error (MSE) are used as available losses. The results indicate that representing angles using trigonometric functions yields improved performance, which is consistent with the

TABLE VII

VERIFICATION PERFORMANCE (%) ON PARTIAL FINGERPRINTS FROM PCF. THE GROUPING RULES ARE THE SAME AS TABLE VI

Matcher	Method	EER	FNMR ¹	FNMR ²
VeriFinger [56]	No Registration	5.69	17.51	33.40
	JointNet [9]	5.90	16.53	33.65
	GridNet [10]	7.21	18.77	35.31
	Cap-MLP [23]	8.18	20.33	36.75
	Cap-CNN [24], [25]	7.85	19.72	36.19
	DRACO (ours)	4.09	12.07	30.08
FDD [15]	No Registration	45.62	89.55	95.04
	JointNet [9]	34.08	82.27	92.66
	GridNet [10]	41.27	89.65	95.50
	Cap-MLP [23]	44.01	91.53	96.80
	Cap-CNN [24], [25]	43.81	91.31	96.15
	DRACO (ours)	17.92	56.01	76.54

¹ FNMR@FMR=1e-3, ² FNMR@FMR=1e-4.

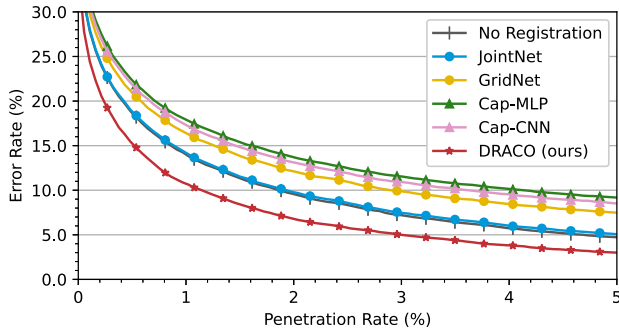


Fig. 11. The VeriFinger [56] based fingerprint indexing performance with corresponding pose constraint on partial fingerprints from PCF. Different input modalities are distinguished by the shape of markers.

TABLE VIII

ABLATION STUDY (°) OF POSE REPRESENTATION FORM ON DPF (FULL FINGERPRINTS WITH ROTATION RANGE OF $[-180^\circ, 180^\circ]$). ALL ABBREVIATIONS ARE DEFINED IN SECTION V-E

Loss\Head	Regressor		Classifier	
	ang	tan	max	sum
MSE (ang)	13.2	7.9	-	7.3
MSE (tan)	12.3	7.1	-	3.3
JS	\	\	2.2	2.0
CE	\	\	2.0	1.8

'\ ' indicates that corresponding item is not applicable.

'-' indicates that corresponding process does not converge.

explanation in Section III-C. On this basis, we further transform the task into probability estimation and explore two classification heads for prediction: (1) *max*: the highest response across all categories is directly selected as the result, and (2) *sum*: the quantized probability distribution and class embedding are weighted and summed as Equation 3. Two classic distribution metrics, cross entropy (CE) and Jensen-Shannon divergence (JS), are supplemented as candidate supervisors. The comparison strongly demonstrates the superiority of our proposed pose representation of probability distribution form and verifies the rationality of the analysis in Section III-C.

TABLE IX

ABLATION STUDY OF FEATURE EXTRACTION AND FUSION ON DPF (PARTIAL FINGERPRINTS WITH ROTATION RANGE OF $[-180^\circ, 180^\circ]$). THE GRAY BACKGROUND INDICATES THE OPTIMAL STRATEGY COMBINATION WITHIN EACH GROUP, WHICH SERVES AS THE BASIS FOR THE FOLLOWING GROUPS

Used Modal	Ridge	Cap	Fusion Strategy [†]	Knowledge Transfer	Avg. Error	
					trans (px)	rot (°)
✓			\	\	25.4	14.1
	✓		\	\	75.7	68.1
✓	✓		\	\	23.5	8.7
✓	✓		E. MoE	\	22.4	7.7
✓	✓		F. MoE	\	22.9	7.3
✓	✓		A. MoE	\	21.3	6.8
✓	✓		A. MoE	Feature	21.0	6.5
✓	✓		A. MoE	Response	21.9	6.8
✓	✓		A. MoE	Relation	19.8	5.5

[†] Abbreviations 'E.', 'F.' and 'A.' respectively represent Equal, Fixed-Weight and Adaptive.

3) *Feature Extraction & Fusion*: Furthermore, the mechanisms and modules proposed in this paper for modal fusion and feature extraction enhancement are verified in Table IX. In the first group of experiments, the significant complementarity between ridge patches and capacitive images is reconfirmed. On this basis, we compare different fusion strategies in the second group. Three typical fusion schemes [58], [59] are evaluated, including (1) *E. MoE*: treating each expert as equally important, with their results directly summed up, (2) *F. MoE*: establishing a set of learnable parameters as globally fixed weights assigned to the experts, (3) *A. MoE*: dynamically assigning sample-specific adaptive weights to different experts through the router depicted in Fig. 2. Strategy (3) outperforms the others, which is understandable as it clearly exhibits the highest flexibility and generalization ability.

The aim of the last comparison group is to assess the effectiveness of knowledge transfer concept introduced in Section III-D. The supervision scheme based on features [60] and responses [61] is additionally examined. From the third and fourth lines to the bottom, it can be seen that directly approximating numerical values from the feature space proves ineffective, likely due to the significant domain difference between full and partial fingerprints. The response based approach even has adverse effects. A convincing explanation is that it disrupts the probability distribution introduced by Equation 6, which may lead to inaccurate representations. Finally, knowledge transfer based on comparative relationships highlights the structured connections between samples, effectively incorporating higher-level semantic information and enhancing model performance.

F. Efficiency

Model size and inference speed of different fingerprint pose estimation algorithms on PCF are summarized in Table X. The time covers a complete process from inputting a sample to outputting the corresponding pose information, which is measured on a single NVIDIA GeForce RTX 3090 GPU by setting the batch size to 1, with an Intel Xeon E5-2680 v4

TABLE X

MODEL SIZE AND AVERAGE TIME COST OF DIFFERENT FINGERPRINT POSE ESTIMATION ALGORITHMS WHEN PROCESSING PCF. METHODS IN EACH GROUP USE DIFFERENT MODAL INPUT

Method	Param (M)	Times (ms)
JointNet [9]	5.30	19.7
GridNet [10]	14.2	16.8
Cap-MLP [23]	0.17	3.6
Cap-CNN [24], [25]	1.70	4.4
DRACO	9.65	26.1

CPU @ 2.4 GHz. All algorithms are implemented in Python (Pytorch). It can be seen that our method exhibits comparable efficiency while delivering high estimation performance, thereby highlighting its attractive practical value.

VI. CONCLUSION

In this paper, we propose DRACO, a novel partial fingerprint pose estimation method under dual-modal collaborative guidance of ridge patches and capacitive images, which are captured by under-screen fingerprint sensors and touch sensors of smartphones. Unlike previous single modal based approaches, we demonstrate the strong complementarity between these two modalities and present an effective framework to integrate and leverage their combined strengths. Specifically, relationship based knowledge transfer and MoE strategies are employed to enhance the network's feature extraction and fusion capabilities. Furthermore, we reformulate fingerprint pose representation into a decoupled probability distribution, significantly improving prediction accuracy. Extensive experiments on multiple databases demonstrate that DRACO outperforms state-of-the-art methods in both precision and robustness. Future work will investigate deeper integration of fingerprint pose estimation with other related tasks and downstream processes, particularly improving its synergy with feature extraction and matching algorithms.

REFERENCES

- [1] A. K. Jain, S. Prabhakar, L. Hong, and S. Pankanti, "Filterbank-based fingerprint matching," *IEEE Trans. Image Process.*, vol. 9, no. 5, pp. 846–859, May 2000.
- [2] K. Nilsson and J. Bigun, "Localization of corresponding points in fingerprints by complex filtering," *Pattern Recognit. Lett.*, vol. 24, no. 13, pp. 2135–2144, Sep. 2003.
- [3] R. Cappelli and D. Maltoni, "On the spatial distribution of fingerprint singularities," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 4, pp. 742–748, Apr. 2009.
- [4] S. Yoon, K. Cao, E. Liu, and A. K. Jain, "LFIQ: Latent fingerprint image quality," in *Proc. IEEE 6th Int. Conf. Biometrics, Theory, Appl. Syst. (BTAS)*, Sep. 2013, pp. 1–8.
- [5] X. Yang, J. Feng, and J. Zhou, "Localized dictionaries based orientation field estimation for latent fingerprints," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 5, pp. 955–969, May 2014.
- [6] J. Ouyang, J. Feng, J. Lu, Z. Guo, and J. Zhou, "Fingerprint pose estimation based on faster R-CNN," in *Proc. IEEE Int. Joint Conf. Biometrics*, Oct. 2017, pp. 268–276.
- [7] C. Deerada, K. Phromsuthirak, A. Rungchokanun, and V. Areekul, "Progressive focusing algorithm for reliable pose estimation of latent fingerprints," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 1232–1247, 2020.
- [8] J. J. Engelsma, K. Cao, and A. K. Jain, "Learning a fixed-length fingerprint representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 6, pp. 1981–1997, Jun. 2021.
- [9] Q. Yin, J. Feng, J. Lu, and J. Zhou, "Joint estimation of pose and singular points of fingerprints," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 1467–1479, 2021.
- [10] Y. Duan, J. Feng, J. Lu, and J. Zhou, "Estimating fingerprint pose via dense voting," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 2493–2507, 2023.
- [11] D. Maltoni, D. Maio, A. K. Jain, and J. Feng, *Handbook of Fingerprint Recognition*. Cham, Switzerland: Springer, 2022.
- [12] K. Cao and A. K. Jain, "Automated latent fingerprint recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 4, pp. 788–800, Apr. 2019.
- [13] X. Guan, Y. Duan, J. Feng, and J. Zhou, "Regression of dense distortion field from a single fingerprint image," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 4377–4390, 2023.
- [14] S. A. Grosz and A. K. Jain, "AFR-Net: Attention-driven fingerprint recognition network," *IEEE Trans. Biometrics, Behav., Identity Sci.*, vol. 6, no. 1, pp. 30–42, Jan. 2024.
- [15] Z. Pan, Y. Duan, J. Feng, and J. Zhou, "Fixed-length dense descriptor for efficient fingerprint matching," in *Proc. IEEE Int. Workshop Inf. Forensics Security*, Dec. 2024, pp. 1–6.
- [16] R. Cappelli, M. Ferrara, and D. Maltoni, "Minutia cylinder-code: A new representation and matching technique for fingerprint recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 12, pp. 2128–2141, Dec. 2010.
- [17] Y. Su, J. Feng, and J. Zhou, "Fingerprint indexing with pose constraint," *Pattern Recognit.*, vol. 54, pp. 1–13, Jun. 2016.
- [18] K. Cao and A. K. Jain, "Fingerprint indexing and matching: An integrated approach," in *Proc. IEEE Int. Joint Conf. Biometrics (IJCBC)*, Oct. 2017, pp. 437–445.
- [19] S. Gu, J. Feng, J. Lu, and J. Zhou, "Latent fingerprint indexing: Robust representation and adaptive candidate list," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 908–923, 2022.
- [20] S. A. Grosz and A. K. Jain, "Latent fingerprint recognition: Fusion of local and global embeddings," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 5691–5705, 2023.
- [21] Q. Yin, J. Feng, J. Lu, and J. Zhou, "Orientation field estimation for latent fingerprints by exhaustive search of large database," in *Proc. IEEE 9th Int. Conf. Biometrics Theory, Appl. Syst. (BTAS)*, Oct. 2018, pp. 1–9.
- [22] S. Gu, J. Feng, J. Lu, and J. Zhou, "Efficient rectification of distorted fingerprints," *IEEE Trans. Inf. Forensics Security*, vol. 13, no. 1, pp. 156–169, Jan. 2018.
- [23] R. Xiao, J. Schwarz, and C. Harrison, "Estimating 3D finger angle on commodity touchscreens," in *Proc. Int. Conf. Interact. Tabletops Surf.*, 2015, pp. 47–50.
- [24] S. Mayer, H. V. Le, and N. Henze, "Estimating the finger orientation on capacitive touchscreens using convolutional neural networks," in *Proc. ACM Int. Conf. Interact. Surf. Spaces*, Oct. 2017, pp. 220–229.
- [25] K. He, C. Li, Y. Duan, J. Feng, and J. Zhou, "TrackPose: Towards stable and user adaptive finger pose estimation on capacitive touchscreens," *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 7, no. 4, pp. 1–22, Dec. 2023.
- [26] Y. Li et al., "TokenPose: Learning keypoint tokens for human pose estimation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 11293–11302.
- [27] J. Li et al., "Human pose regression with residual log-likelihood estimation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 11025–11034.
- [28] Y. Li et al., "SimCC: A simple coordinate classification perspective for human pose estimation," in *Proc. Eur. Conf. Comput. Vis.*, 2021, pp. 89–106.
- [29] T. Jiang et al., "RTMPose: Real-time multi-person pose estimation based on MMPose," 2023, *arXiv:2303.07399*.
- [30] K. C. Mangold, "Data format for the interchange of fingerprint, facial & other biometric information ANSI/NIST-ITL 1–2011 NIST special publication 500–290 edition 3," Nat. Inst. Standards Technol., Gaithersburg, MD, USA, Tech. Rep. NIST SP 500-290e3, 2016.
- [31] G. T. Candela, P. J. Grother, C. I. Watson, R. A. Wilkinson, and C. L. Wilson, *PCASYS—A Pattern-Level Classification Automation System for Fingerprints*, Standard NISTIR 5647, 1995.
- [32] C. Watson, P. Flanagan, and B. Cochran, "Slap fingerprint segmentation evaluation II (slapsegII)," Dept. U.S. Dept. of Commerce, Nat. Inst. Standards Technol., Gaithersburg, MD, USA, Tech. Rep. 7553, 2009.
- [33] A. M. Bazen and S. H. Gerez, "Segmentation of fingerprint images," in *Proc. Workshop Circuits, Syst. Signal Process. (ProRISC)*, 2001, pp. 276–280.
- [34] M. Liu, X. Jiang, and A. C. Kot, "Fingerprint reference-point detection," *EURASIP J. Adv. Signal Process.*, vol. 2005, no. 4, pp. 1–12, Dec. 2005.

- [35] K. Rerkrai and V. Areekul, "A new reference point for fingerprint recognition," in *Proc. Int. Conf. Image Process. (ICIP)*, vol. 2, Sep. 2000, pp. 499–502.
- [36] V. Areekul, K. Suppasriwasuth, and S. Jirachaweng, "The new focal point localization algorithm for fingerprint registration," in *Proc. Int. Conf. Pattern Recognit. (ICPR)*, vol. 4, Aug. 2006, pp. 497–500.
- [37] X. Si, J. Feng, J. Zhou, and Y. Luo, "Detection and rectification of distorted fingerprints," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 555–568, Mar. 2015.
- [38] P. Schuch, J. M. May, and C. Busch, "Unsupervised learning of fingerprint rotations," in *Proc. Int. Conf. Biometrics Special Interest Group (BIOSIG)*, Sep. 2018, pp. 1–6.
- [39] G. Arora, A. Kumbhat, A. Bhatia, and K. Tiwari, "CP-Net: Multi-scale core point localization in fingerprints using hourglass network," in *Proc. 11th Int. Workshop Biometrics Forensics (IWBf)*, Apr. 2023, pp. 1–6.
- [40] S. A. Grosz, J. J. Engelsma, E. Liu, and A. K. Jain, "C2CL: Contact to contactless fingerprint matching," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 196–210, 2022.
- [41] Z. He, J. Zhang, L. Pang, and E. Liu, "PFVNet: A partial fingerprint verification network learned from large fingerprint matching," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 3706–3719, 2022.
- [42] X. Guan, Z. Pan, J. Feng, and J. Zhou, "Joint identity verification and pose alignment for partial fingerprints," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 249–263, 2025.
- [43] V. Zaliva, "3D finger posture detection and gesture recognition on touch surfaces," in *Proc. 12th Int. Conf. Control Automat. Robot. Vis. (ICARCV)*, Guangzhou, China, Dec. 2012, pp. 359–364.
- [44] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5987–5995.
- [45] S. Woo, J. Park, J. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.
- [46] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural Comput.*, vol. 3, no. 1, pp. 79–87, Mar. 1991.
- [47] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *J. Mach. Learn. Res.*, vol. 23, no. 120, pp. 1–39, 2022.
- [48] A. Liu et al., "DeepSeek-V3 technical report," 2024, *arXiv:2412.19437*.
- [49] H. Touvron et al., "ResMLP: Feedforward networks for image classification with data-efficient training," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 5314–5321, Apr. 2023.
- [50] J.-B. Grill et al., "Bootstrap your own latent—A new approach to self-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21271–21284.
- [51] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.
- [52] S. Zhou, W. Liu, C. Hu, S. Zhou, and C. Ma, "UniDistill: A universal cross-modality knowledge distillation framework for 3D object detection in bird's-eye view," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5116–5125.
- [53] C. Yang et al., "CLIP-KD: An empirical study of CLIP model distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 15952–15962.
- [54] *FVC 2002: The Second International Competition for Fingerprint Verification Algorithms*. Accessed: Apr. 17, 2025. [Online]. Available: <http://bias.csr.unibo.it/fvc2002/default.asp>
- [55] *FVC 2004: The Third International Fingerprint Verification Competition*. Accessed: Apr. 17, 2025. [Online]. Available: <http://bias.csr.unibo.it/fvc2004/default.asp>
- [56] Neurotechnology.(2024). *VeriFinger SDK*. [Online]. Available: <https://www.neurotechnology.com/verifinger.html>
- [57] Y. Duan, J. Yu, J. Feng, K. He, J. Lu, and J. Zhou, "3D finger rotation estimation from fingerprint images," *Proc. ACM Hum.-Comput. Interact.*, vol. 7, no. ISS, pp. 114–134, Oct. 2023.
- [58] Y. Wang, "Survey on deep multi-modal data analytics: Collaboration, rivalry, and fusion," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 17, no. 1s, pp. 1–25, Jan. 2021.
- [59] F. Zhao, C. Zhang, and B. Geng, "Deep multimodal data fusion," *ACM Comput. Surveys*, vol. 56, no. 9, pp. 1–36, Oct. 2024.
- [60] J. Li et al., "Rethinking feature-based knowledge distillation for face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 20156–20165.
- [61] C. Yang, X. Yu, Z. An, and Y. Xu, *Categories of Response-Based, Feature-Based, and Relation-Based Knowledge Distillation*. Cham, Switzerland: Springer, 2023, pp. 1–32.



Xiongjun Guan received the B.Eng. degree from the Department of Automation, Tsinghua University, Beijing, China, in 2021, where he is currently pursuing the Ph.D. degree, under the supervision of Prof. Jianjiang Feng. His research interests include computer vision, pattern recognition, and human–computer interaction.



Zhiyu Pan received the B.Eng. degree in electronic science and technology from Beijing Institute of Technology, China, in 2020. He is currently pursuing the Ph.D. degree with the Department of Automation, Tsinghua University. His research interests include biometrics, human action analysis, and computer vision. Specifically, his current work focuses on fingerprint recognition, multi-modal learning, and related areas.



Jianjiang Feng (Member, IEEE) received the B.Eng. and Ph.D. degrees from the School of Telecommunication Engineering, Beijing University of Posts and Telecommunications, China, in 2000 and 2007, respectively. From 2008 to 2009, he was a Post-Doctoral Researcher with the PRIP Laboratory, Michigan State University. He is currently an Associate Professor with the Department of Automation, Tsinghua University, Beijing. His research interests include fingerprint recognition and computer vision.



Jie Zhou (Fellow, IEEE) received the B.S. and M.S. degrees from the Department of Mathematics, Nankai University, Tianjin, China, in 1990 and 1992, respectively, and the Ph.D. degree from the Institute of Pattern Recognition and Artificial Intelligence, Huazhong University of Science and Technology (HUST), Wuhan, China, in 1995. Since 1997, he has been a Post-Doctoral Fellow with the Department of Automation, Tsinghua University, Beijing, China. Since 2003, he has been a Full Professor with the Department of Automation, Tsinghua University.

In recent years, he has authored more than 300 papers in peer-reviewed journals and conferences. Among them, more than 100 papers have been published in top journals and conferences, such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON IMAGE PROCESSING, and CVPR. His research interests include computer vision, pattern recognition, and image processing. He is an IAPR Fellow. He received the National Outstanding Youth Foundation of China Award. He is an Associate Editor of IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE and two other journals.